

CoSolutioning

The Human + AI Collaboration Framework for building the next generation of AI-First/AI-Native organizations, products, and services

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We are at a moment when our ability to create solutions is changing radically. Ways of working have always evolved in response to the tools and capabilities available at different points in history. Artificial Intelligence represents one of those turning points: a capability that learns, reasons, creates, executes, and coordinates work at an exponentially increasing speed.

Yet something essential remains unchanged: we create value when we solve real problems that people face—and when we do so in meaningful ways. Building a solution continues to make sense when it creates value for people and, as a result, generates value for the business. **That is why we preserve principles rooted in the scientific method:** observing reality, understanding the problem, formulating hypotheses, experimenting, measuring evidence, learning, and evolving. Technology accelerates our ability to act; evidence continues to guide what we build and how it should evolve.

Applying the foundations of the scientific method enables us to address a fundamental reality: the nature of how the world works and how people behave.

AI is evolving at an exponential pace, while organizations tend to transform gradually. This difference in speed presents one of today's greatest challenges: designing organizations capable of evolving at the pace of this new context. This requires rethinking teams, roles, structures, and ways of working to integrate AI capabilities and turn them into an organizational capability.

This leads us to view AI as part of organizational design. Today, a small human team can lead a coordinated ecosystem of AI systems capable of contributing to the discovery, delivery, and evolution of solutions. The knowledge and capabilities available to a team are no longer limited exclusively by the number of people on it. The right combination of human intelligence + artificial intelligence + cross-functional organizational knowledge can radically multiply what a team is capable of achieving.

AI moves beyond the role of an assistant and becomes an active member of the team, contributing knowledge, skills, and capabilities that, when combined with human strengths, expand the boundaries of what we can imagine, build, and accomplish together.

In this framework, we refer to teams made up of human and artificial intelligences as “Solution Teams.”

This framework recognizes that leadership, judgment, and accountability remain human responsibilities. Ethical, moral, and strategic decisions are entrusted to people. Therefore, every AI system has a designated human owner who provides context and judgment, defines its objectives, and assumes responsibility for its performance and the outcomes it produces.

We are entering an era in which our ability to act can grow exponentially, yet creating value will continue to depend on our ability to choose wisely what to pursue. AI expands the boundaries of what is possible; purpose, human judgment, and evidence help us decide what is worth making possible. Together, we will solve real problems and create value for people.

Principles

- We lead with purpose, understanding the fundamentals of what we do.
- We connect the business, its strategy, and the end user, creating value for both.
- Knowledge must be available where value is created, preventing specialization from becoming a dependency.
- AI is a full member of the team, amplifying its collective capacity to create value.
- We design teams around the capabilities they need to solve problems, regardless of whether those capabilities come from a person, an AI system, or an organizational design that enables cross-functional collaboration.
- Knowledge and information must be accessible and understandable to both the people and the AI systems that are part of the team.
- Events establish rhythms and habits while providing a clear structure for creating value.
- We continuously evolve Human + AI collaboration to strengthen its collective capacity to create value.
- Humans remain at the center—both employees and customers.
- Fostering critical thinking is essential.

- The power of simplicity: Simplicity creates clarity, clarity enables action, and sustained action creates value. What is simple is understood, executed, and turned into value more quickly.
- Simplifying does not mean trivializing. Simplification is the art of identifying the parts of a complex system and addressing them in a way that preserves the meaning and integrity of the system as a whole.
- Solution Teams conduct Discovery and Delivery simultaneously, answering two essential questions:
Discovery: Are we solving the right problem?
Delivery: Are we building and evolving the solution in the right way?

This framework is a system of principles, concepts, elements, and practices that guides how organizations and teams make decisions and take action within the emerging landscape of Human + AI collaboration. It is important to emphasize that the framework establishes a structure within which organizations can make decisions and adapt their ways of working to their specific context.

The framework operates through the following elements:

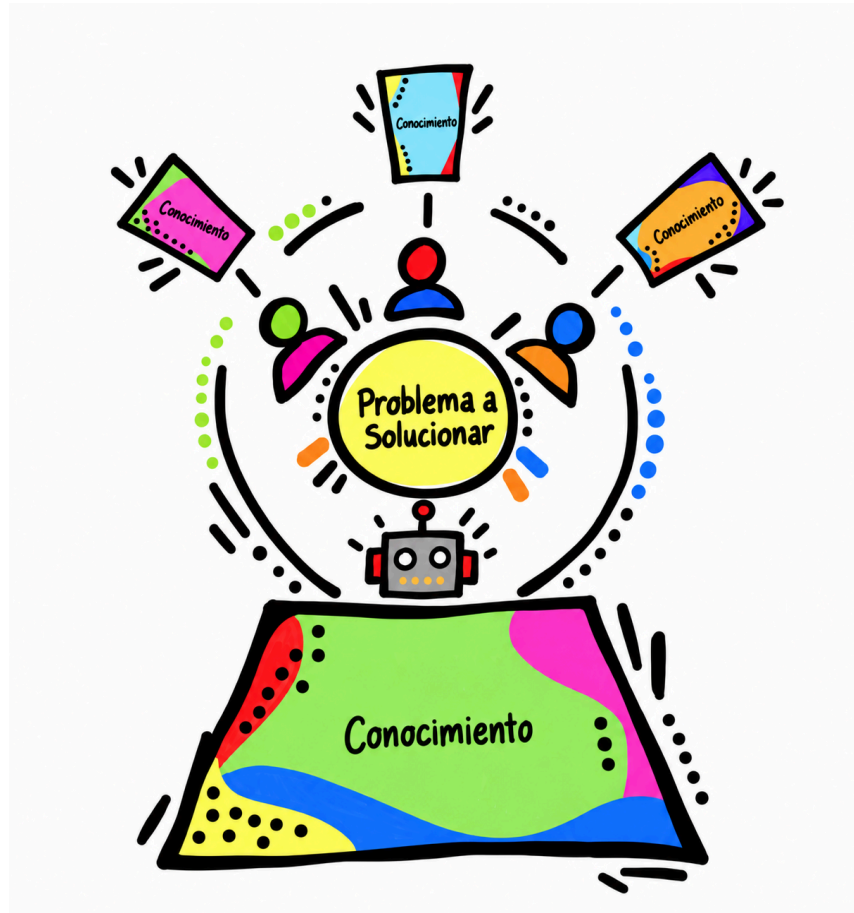
Roles

The essential goal is to ensure that every team has the knowledge and capabilities required to create and evolve end-to-end solutions. These capabilities emerge from the combination of human talent, Artificial Intelligence, and the specialized knowledge of the organization's cross-functional teams. Teams are multidisciplinary and bring together the skills needed to create value for both users and the business, integrating Discovery and Delivery within the same team.

This composition fosters autonomy, reduces dependencies, and enables teams to continuously connect learning with the development and evolution of solutions.

We begin with a fundamental idea: a team consists of a group of people who, through their different roles, contribute complementary knowledge, experience, skills, and capabilities to solve problems and build solutions. The team's strength comes from this combination.

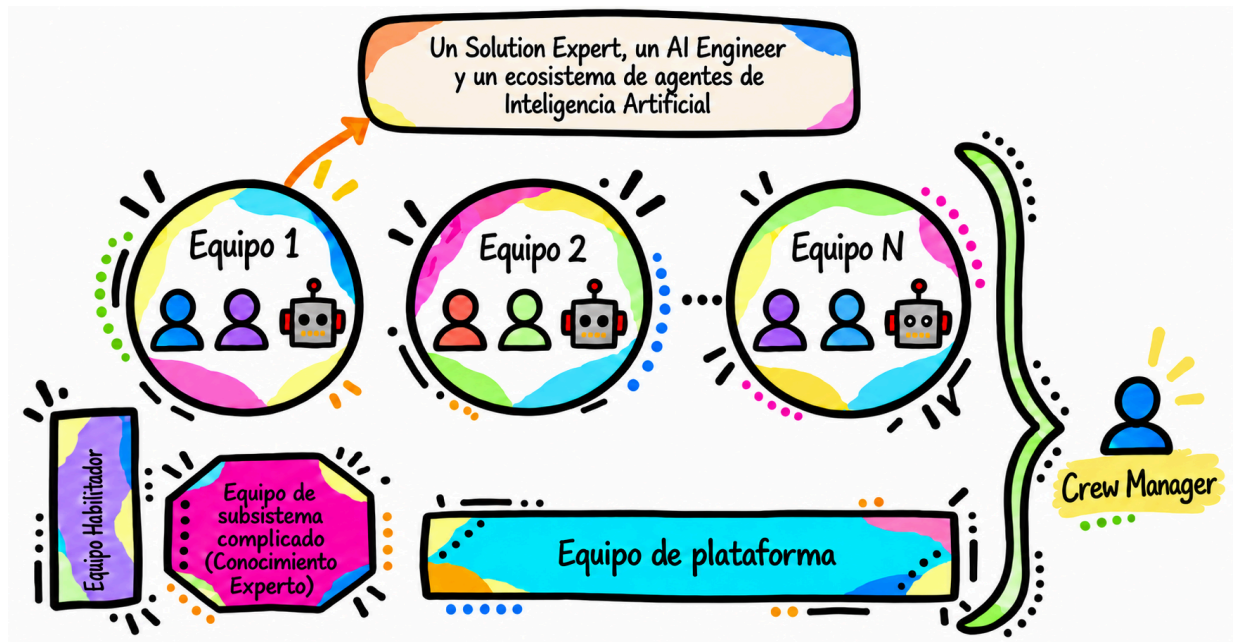
The integration of Artificial Intelligence profoundly transforms this equation. AI moves beyond the role of an assistant and becomes an active member of the team, contributing knowledge, capabilities, skills, and execution capacity. This significantly expands the capabilities available to build digital products and solutions.



Combination of Human + AI knowledge, skills, and capabilities

The result is a team with access to a far broader universe of knowledge and capabilities than any one person could possess. The team's strength increasingly depends on how effectively humans and AI systems combine their capabilities to solve problems and create value together.

This new context transforms how we design teams and define roles. The organization evolves toward a model composed of Human + AI teams, complemented by cross-functional capabilities that provide specialized knowledge.



Organizational design of Human + AI teams and cross-functional capabilities

In this way, knowledge is available where value is created.

Each Solution Team consists of a Solution Expert, an AI Engineer, and an ecosystem of Artificial Intelligence agents. Together, they form an autonomous unit capable of understanding the problem, designing and building the solution, and continuously evolving it.

These capabilities are strengthened by an organizational design based on the principles of [Team Topologies](#). Surrounding the Solution Teams is a network of cross-functional teams that develop and share specialized knowledge, accelerate the adoption of new capabilities, and provide services. This ecosystem consists of Enabling Teams, which help teams acquire new knowledge, capabilities, and skills; Complicated Subsystem Teams, which address challenges requiring deep expertise in highly complex domains; and Platform Teams, which build the platforms and services that enable Solution Teams to create, deploy, and deliver value.

Cross-functional teams and Solution Teams interact in three ways: by facilitating capability development, collaborating directly to address specific challenges, or providing capabilities through an X-as-a-Service model, in which one team offers specialized services that others can consume on demand. The appropriate interaction mode is selected according to the needs and context of each situation.

This model combines human and artificial intelligence. People contribute judgment, context, creativity, objectives, and the purpose behind what we are building, while retaining full ownership and accountability for what is created. AI contributes the knowledge, skills, and capabilities required to build solutions with greater speed and quality.

We believe this organizational design creates the conditions for human intelligence and AI to enhance one another, bringing together all the knowledge, skills, and capabilities required to create solutions that deliver significant value to users and businesses.

When a solution requires multiple Solution Teams to work together, we propose introducing a role called the Crew Manager. This role is responsible for orchestrating coordination, managing dependencies, and maintaining alignment across teams to ensure a continuous flow of value delivery. We recommend activating this role when building and evolving a solution requires the simultaneous participation of three or more teams, as coordination complexity begins to increase significantly at that point.

For solutions involving only one or two teams, we do not consider this role necessary. In these cases, direct communication, shared accountability, and day-to-day collaboration are sufficient to keep teams synchronized without introducing additional layers of coordination.

The roles required within this framework are described below:

- **Solution Expert:**

The Solution Expert is a business domain expert who guides the strategy and evolution of digital solutions through a deep understanding of the business and its users, using Artificial Intelligence to strengthen decision-making.

Key responsibilities:

- ➔ Deep knowledge of the industry, business, and users: Understand the context, regulations, market dynamics, and key problems faced by users, customers, and stakeholders. The Solution Expert draws on AI as a repository of vast amounts of knowledge. However, the ability to manage and validate that knowledge—combined with firsthand experience—is what sets an effective Solution Expert apart.

- Defining objectives and outcomes: Translate the business vision into clear, measurable objectives and outcomes. Maintain an outcome-based solution roadmap supported by OKRs.
- Advanced use of AI: Leverage AI capabilities—such as agents, skills, and MCPs—to explore trends, anticipate customer needs, and accelerate innovation.
- Continuous learning: Continuously develop an understanding of AI capabilities and how to apply them effectively within the role.
- User-centered solution design: Lead the development of products that address meaningful user problems, recognizing users as the people whose problems we seek to solve and for whom we aim to create value.
- Portfolio management: Prioritize initiatives based on evidence, outcomes, and business value.
- Collaborative leadership: Act as a bridge between business, technology, and the market, guiding the team toward a shared purpose.

- **AI Engineer:**

An AI Engineer is a professional who combines software engineering with advanced AI implementation to design, build, and deploy robust, scalable technology solutions that deliver significant business value.

The AI Engineer focuses on the practical application of AI and system architecture: They select, train, optimize, and integrate proprietary or third-party AI models into applications and end products across small-, medium-, and large-scale environments. They also bridge the gap between business needs and technical feasibility, ensuring that the software lifecycle meets rigorous standards for quality, traceability, and ethics.

Key responsibilities:

- Solution architecture and development: Design the technology architecture and integrate AI models into technology solutions.
- Model selection, training, and fine-tuning: Evaluate and test different AI tools and models, optimizing them to address specific needs.

- MLOps, versioning, and traceability: Manage the complete lifecycle of models and applications, including version control, security, monitoring, and production scalability.
- Continuous experimentation and quality: Apply AI-driven experimentation techniques to validate hypotheses iteratively and identify the best functional and technical solution.
- Technical quality: Establish sound development, testing, and monitoring practices to ensure that solutions are robust and secure.
- Governance and ethics: Ensure that the use of AI complies with governance, privacy, organizational architecture, and responsible-use guidelines.
- Strategic collaboration: Work alongside business teams and Solution Experts to translate user needs into technically feasible requirements.
- AI harness architecture and development (Harness Engineering): Design, implement, and maintain the orchestration, evaluation, and control infrastructure—the harness—surrounding large language models (LLMs) and autonomous agents. Ensure that AI systems operate safely and autonomously by integrating guardrails, automated self-correction loops such as linters and unit tests, tooling governance, and Human-in-the-Lead control mechanisms.

Human-in-the-Lead means moving beyond viewing people as checkpoints or passive validators of AI and positioning them as strategic leaders and active decision-makers, with AI serving as an amplifier of their capabilities.

● **Crew Manager:**

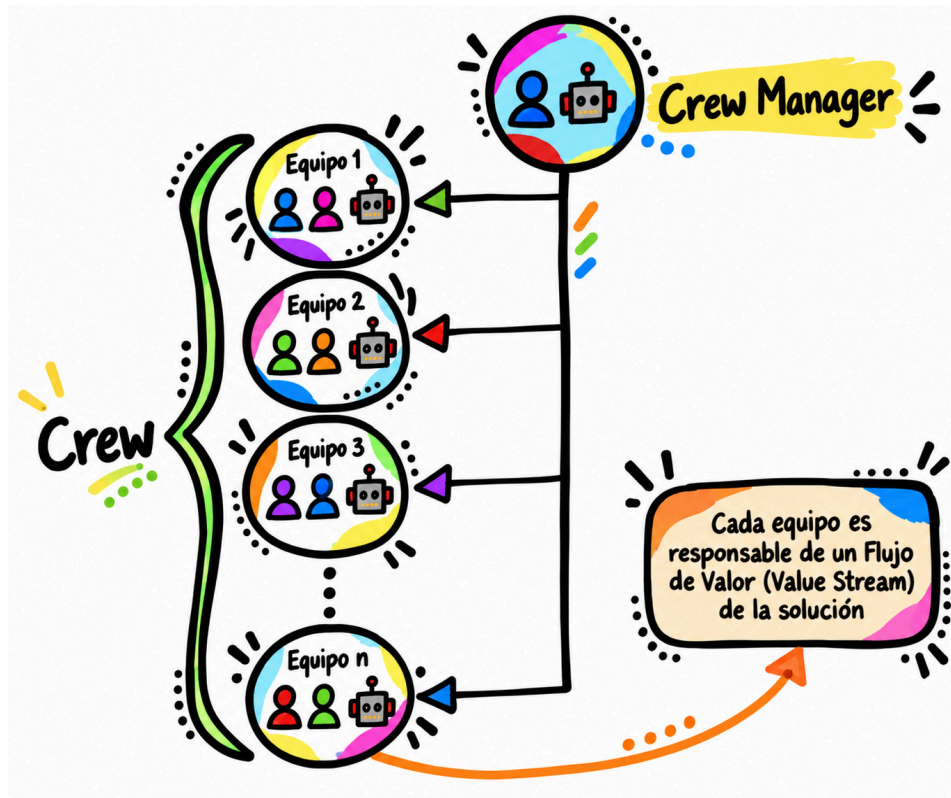
The Crew Manager guides and orchestrates the work of multiple Solution Teams within a scaled model, ensuring strategic alignment, technological synergy, and a clear focus on creating value for both the business and its users. This role connects vision, prioritization, and execution across teams, enabling AI-powered solutions to scale consistently and securely while delivering measurable impact for the organization and its users.

The Crew Manager brings multiple teams together within an integrated ecosystem aligned with the solution's overarching vision and the organization's strategy. They ensure that AI-driven innovation can scale, that teams remain focused on outcomes, and that they work in synergy.

Key responsibilities:

- Solution Team orchestration: Coordinate and synchronize the work of multiple Human + AI teams, manage dependencies, and maintain alignment across teams to ensure a continuous flow of value delivery.
- Alignment with business strategy: Ensure that the initiatives developed by the teams are connected to the organization's strategic objectives.
- Metrics: Help define, monitor, and evolve indicators that measure business value, outcomes, productivity, and team health, enabling evidence-based decision-making.
- Culture of collaboration: Foster an environment of trust, continuous learning, and alignment around a shared purpose that inspires and guides the teams.

The following diagram illustrates how the Crew Manager interacts with three or more teams:



The Crew Manager role, the teams that make up the Crew, and how they interact

We use the term Crew to refer to a group of Solution Teams, as these teams represent the fundamental unit for creating and evolving value within the organization. Each Solution Team is responsible for a specific Value Stream and has the autonomy and capabilities required to understand the problem, discover opportunities, and continuously design, build, deploy, and evolve its solutions based on data, evidence, and learning.

When a solution requires the collaboration of more than three Solution Teams, the Crew Manager assumes responsibility for orchestrating coordination, managing dependencies, and maintaining alignment across teams, ensuring a continuous flow of value delivery. The purpose of this role is to facilitate collaboration in more complex environments where multiple humans and AI systems interact, enabling teams to remain focused on creating value for users and the business.

Note: It is important to emphasize that every Solution Team conducts Discovery and Delivery and continuously evolves its solutions.

Artifacts

Artifacts are designed to build and maintain a shared understanding. They make key information accessible, visible, and easy to understand by bringing together knowledge, evidence, decisions, and learning into a shared source of knowledge for teams. Artifacts enable everyone involved to make better decisions.

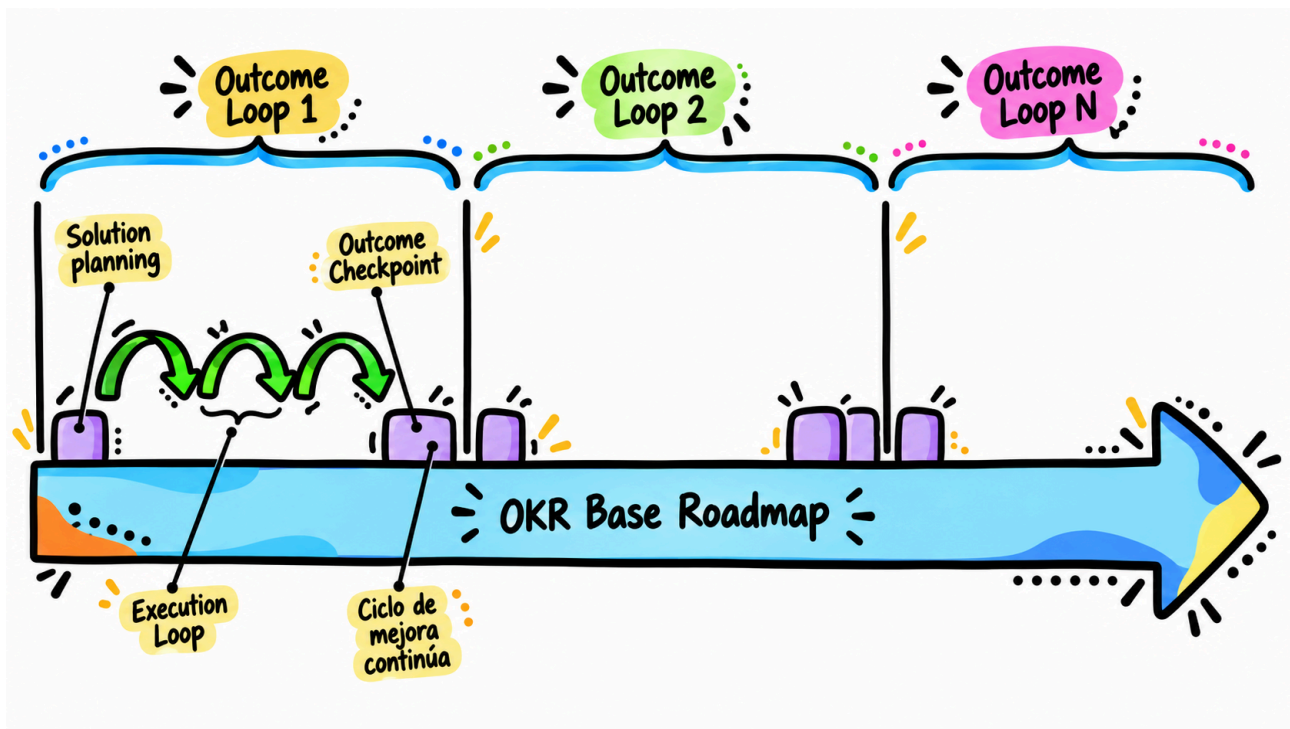
The framework uses the following artifacts:

- A document outlining the problems to be solved and the hypotheses to be validated.
- OKR-Based Roadmaps.
- Version-controlled specification documents: Artifacts that define the product and its requirements.
- Learning Log.
- Harness: A structure containing the rules, guardrails, and automated validation loops that constrain and guide an AI agent's execution to ensure safe, measurable, scalable, and adaptable outcomes.

Events

Events establish a working cadence that helps turn the framework's practices into habits, which is essential to the success of its ways of working. They create recurring opportunities to align, discover, build, review evidence, learn, and adapt the work, providing structure to the continuous cycle of Discovery, Delivery, and solution evolution.

The following diagram presents all the events within the CoSolutioning Framework on a timeline. Each event is described in the sections that follow:



Timeline of Events and the OKR-Based Roadmap

- **Outcome Loop:** The Solution Team's work cycle, bringing together the Solution Expert, the AI Engineer, and their AI ecosystem. It is the period during which the team creates an increment of value and advances toward a defined outcome.

Each team determines the duration of its Outcome Loop, which may range from one to five days. Once established, the team maintains a consistent cadence over time. This regularity supports a sustainable pace of work, facilitates synchronization among team members, and encourages the continuous practice of Discovery, Delivery, solution evolution, and evidence-based continuous improvement.

The cadence of Outcome Loops also encourages interaction among Solution Teams, helping turn this way of working into a habit.

Within each Outcome Loop, the team may complete one or more **Execution Loops**. An Execution Loop is a short execution cycle focused on building, validating, or evolving a specific part of the solution. Each team determines how many Execution Loops to complete based on the context, the complexity of the work, and the team's capacity, while maintaining a continuous focus on creating value throughout the Outcome Loop.

- **Solution Planning:** A collaborative session in which the team develops a shared understanding of the problem to be solved, the OKRs to be achieved, and the related business indicators, and then selects the initiative or initiatives to pursue. The entire team participates in this session once per Outcome Loop, refining the plan as needed.

Solution Planning covers the following activities:

- Discuss the problem to be solved.
 - Review the OKR-Based Roadmap.
 - Review business indicators.
 - Prioritize and select the initiatives to pursue in order to achieve the OKRs.
- **Execution Loop:** The cycle in which the team carries out the work defined during Solution Planning. It focuses on building a feature or prototype from end to end using specifications (Specs), with the goal of making it fully functional.

The Execution Loop includes the following activities:

- Creating artifacts for Human + AI collaboration: The team creates the artifacts required to communicate clearly and precisely to AI what needs to be built. We propose Specs as the shared language between humans and AI systems: a mechanism for making intent explicit, reducing ambiguity, and translating decisions into executable instructions. This enables teams to build solutions in a traceable, reproducible, secure, and adaptable way.

- Generating initiative Specs: The Solution Team creates specifications that clearly describe what is expected to be built and provide the AI ecosystem with the context, rules, and criteria required to execute the development work. Specs are managed as version-controlled artifacts, maintaining traceability of both their evolution and the instructions used for each version of the solution.
- Reviewing and validating Specs: Before development begins, the Solution Team reviews the Specs for quality, clarity, completeness, and consistency, ensuring a shared understanding of the solution.
- Executing Specs: The Solution Team executes the Specs using formats that enable the AI systems within the ecosystem to interpret them effectively, such as Markdown (.md) files. Based on these specifications, the AI systems contribute to the end-to-end development of the digital asset, including the frontend, backend, data layer, and any other technical components required by the solution.
- Human + AI quality control: The Solution Team performs functional and technical testing by combining the specialized capabilities of the AI ecosystem with human judgment and oversight. The resulting findings inform adjustments to both the Specs and the solution, creating rapid cycles of building, validation, and improvement.
- Building quality throughout the cycle: Quality is built throughout the entire cycle rather than assessed through an isolated validation step at the end. The Solution Team, composed of humans and AI systems, continuously validates and refines its understanding of the problem, the OKRs, the initiatives, the specifications, and the resulting solution. Each stage incorporates evidence and learning, enabling the team to identify deviations early, reduce errors, and ensure that what is built remains aligned with the problem and the value it seeks to create.
- Completing the initiative: Development is considered complete when the solution meets the established quality and acceptance criteria, preserves traceability between the Specs and the implementation, and is ready to be deployed, used, and subsequently evolved based on evidence.
- Aligning Specs with the Harness: Specs are built on the Harness established for the project, ensuring that requirements comply with its

rules, tools, and constraints from the outset. The AI integrated into the framework acts as a facilitator, guiding the co-creation and automated validation of these specifications.

- **Outcome Checkpoint:** This event takes place at the end of each Outcome Loop and brings together the Solution Team and the stakeholders involved in the initiative. The Outcome Checkpoint covers the following activities:
 - Reviewing the OKR-Based Roadmap.
 - Reviewing business indicators.
 - Inspecting the initiatives that have been built.
 - Sharing the learning generated during the loop.

The [OKR-Based Roadmap](#) is a living artifact that evolves continuously as the team gains new insights and gathers evidence through its interactions with users and the business. In this sense, the OKR-Based Roadmap is dynamic rather than a static plan. It makes it possible to visualize cumulative progress toward the OKRs, adjust priorities, and make evidence-based decisions throughout the Discovery and value-delivery process.

- **Continuous Improvement Cycle:** Its purpose is to increase the effectiveness of Solution Teams by improving Human-to-Human, Human-to-AI, and AI-to-AI working dynamics. Based on evidence of the Solution Team's performance, the team identifies and implements improvements to its practices, interactions, human capabilities, AI capabilities, and collaboration methods to enhance quality, speed, and the value created.

The team discusses and decides:

- How to improve the effectiveness of human contributions.
- How to improve the effectiveness of AI contributions.
- How to improve the effectiveness of Human + AI collaboration.

Authors



Jonnathan Bermudez

With more than 20 years of experience creating solutions and connecting business needs with the possibilities of technology. His work integrates strategy, technology architecture, organizational design, and artificial intelligence capabilities to transform real challenges into useful, measurable solutions capable of evolving. He drives new forms of collaboration between people and AI, promoting an intentional integration in which purpose, human judgment, and accountability guide technology decisions. His contribution to the framework helps organizations incorporate AI into their ways of working in a practical, sustainable way focused on generating value for people and the business.



Felipe Soto

Consultant, speaker, and facilitator, dedicated to helping people and organizations develop their potential through human-centered innovation and leadership. With more than 10 years of experience across industries, he has supported teams in connecting empathy, strategy, and execution to generate meaningful impact. He believes that sustainable transformations occur when organizations cultivate a service-driven leadership culture, where leaders grow by helping others develop and reach their full potential. His work combines the rigor of Lean and Agile practices with the creativity of Design Thinking, always connected to real-world challenges and opportunities.

**Cristian Bermudez**

Professional with more than 19 years of experience leading teams in the construction of high-impact solutions and guiding organizations through transformation processes. He brings deep knowledge of methodologies, work dynamics, and organizational design focused on generating value, connecting strategy, people, and execution. His contribution to the framework makes it possible to design new forms of collaboration between people and AIs, intentionally integrating their capabilities and creating dynamics that enhance what they can achieve together. He keeps the focus on generating value for people and the business, connecting purpose with solutions that can be built, measured, validated, and evolved.

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